

Review of Important points from last lecture

- consider a any physical system with energy $E(\vec{s})$ at temperature T
- suppose we want to obtain cumulants of some observable $\Theta(\vec{s})$
- turn on an external field that lowers energy of states $\vec{s}$ in proportion to $\Theta(\vec{s})$

ie $E(\vec{s}) \rightarrow E(\vec{s}, h) \equiv E(\vec{s}) - kT h \Theta(\vec{s})$

- compute free energy of this new system

$$F(\beta, h) = -kT \ln \int d\vec{s} e^{-\beta E(\vec{s}, h)}$$

- $C_n(\theta) = \left. \frac{\partial^n}{\partial h^n} [-\beta F(\beta, h)] \right|_{h=0}$ - $\beta F(\beta, h)$ is a cumulant generating function for θ

- $\left. \begin{array}{l} C_1(\theta) = \langle \theta \rangle \\ C_2(\theta) = \text{var}(\theta) \end{array} \right\} \Rightarrow \left. \frac{\partial \langle \theta \rangle}{\partial h} = \text{var}(\theta) \right|_{h=0}$ Fluctuation response theorem.
response fluctuation

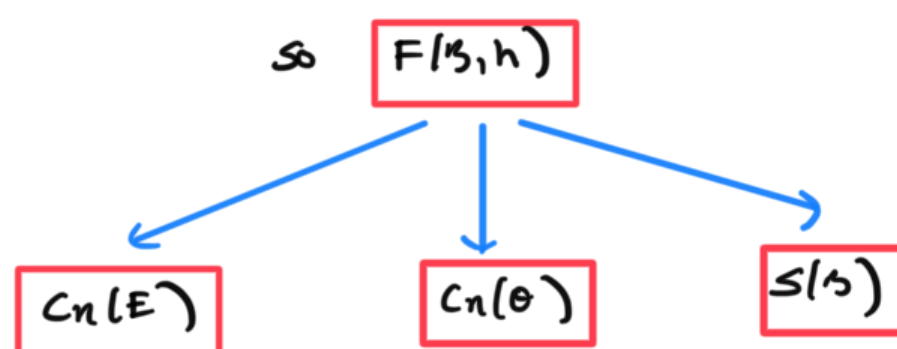
- also when $h=0$

$$C_n(E) = (-1)^n \frac{\partial^n}{\partial \beta^n} [-\beta F(\beta)]$$

- $\beta F(\beta)$ as cumulant gen function for energy...

- also

$$S(\beta) = \beta [\partial_\beta (\beta F) - F(\beta)]$$



Free energy as the source of all information...

From microscopic to macroscopic through a saddle point

$$-\beta F(\beta) = \ln \int d\vec{s} e^{-\beta E(\vec{s})}$$

$$= \ln \int dE e^{-\beta E} \int d\vec{s} \delta[E(\vec{s}) - E]$$

$$= \ln \int dE e^{-\beta E} \Omega(E) e^{-\beta [E - kT S(E)]}$$

$\Omega(E) = e^{S(E)}$

$$= \ln \int dE e^{-\beta N [\epsilon - kT S(\epsilon)]}$$

$$-\beta F(\beta) = \ln \int dE e^{-\beta N [\epsilon - kT S(\epsilon)]}$$

often energy and entropy are
extensive $\propto N$ # degrees of freedom

$$E \equiv N \epsilon \quad \epsilon = \frac{E}{N} \text{ energy density}$$

$$S(E) \equiv N s(\epsilon)$$

$$F(\beta) \equiv N f(\beta)$$

as $N \rightarrow \infty$

$$-\beta N [\epsilon^* - kT S(\epsilon^*)]$$

$$-\beta F(\beta) \approx \ln e$$

$$f(\beta) \approx \epsilon^* - kT S(\epsilon^*)$$

$$\epsilon^* = \operatorname{argmin} [\epsilon - kT S(\epsilon)]$$

$$1 - kT \frac{\partial S}{\partial \epsilon} = 0$$

$$\frac{1}{kT} = \frac{\partial S}{\partial \epsilon}$$

Note we can write this symmetrically as

$$\beta F(\beta) + S(\epsilon^*) = \beta \epsilon^*$$

$$\text{where } \epsilon^* \text{ obeys } \beta = \frac{\partial S}{\partial \epsilon}$$

$\Rightarrow \beta F(\beta)$ and $S(\epsilon)$ are Legendre dual

Review Legendre duality

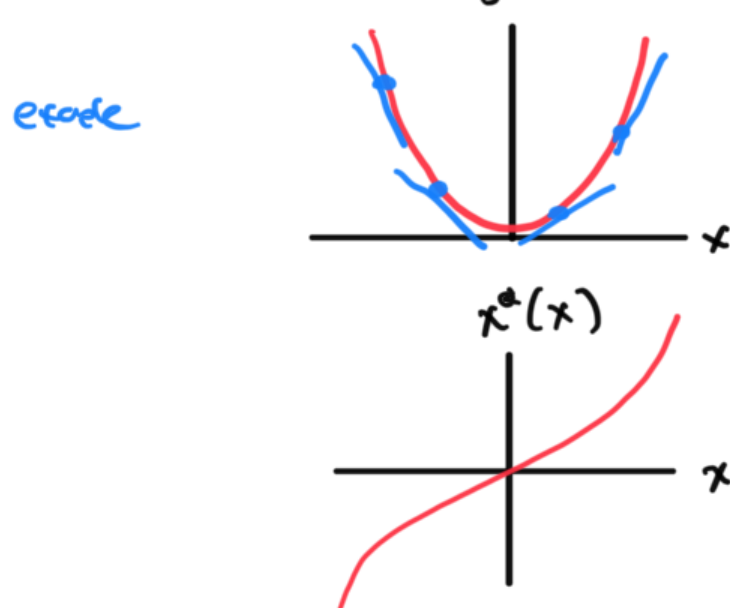
- given a function $y(x)$ that is either strictly convex or concave
ie $\frac{\partial^2 y}{\partial x^2} > 0$ $\frac{\partial^2 y}{\partial x^2} < 0$

- now if $\frac{\partial y}{\partial x}$ is easier to measure or control or think about

- Legendre transform provides a nice way to encode info in $y(x)$

- let $x^*(x) \equiv \frac{\partial y}{\partial x}$ = slope of y at x

- strict convexity/concavity $\Rightarrow x^*(x)$ is one to one map...

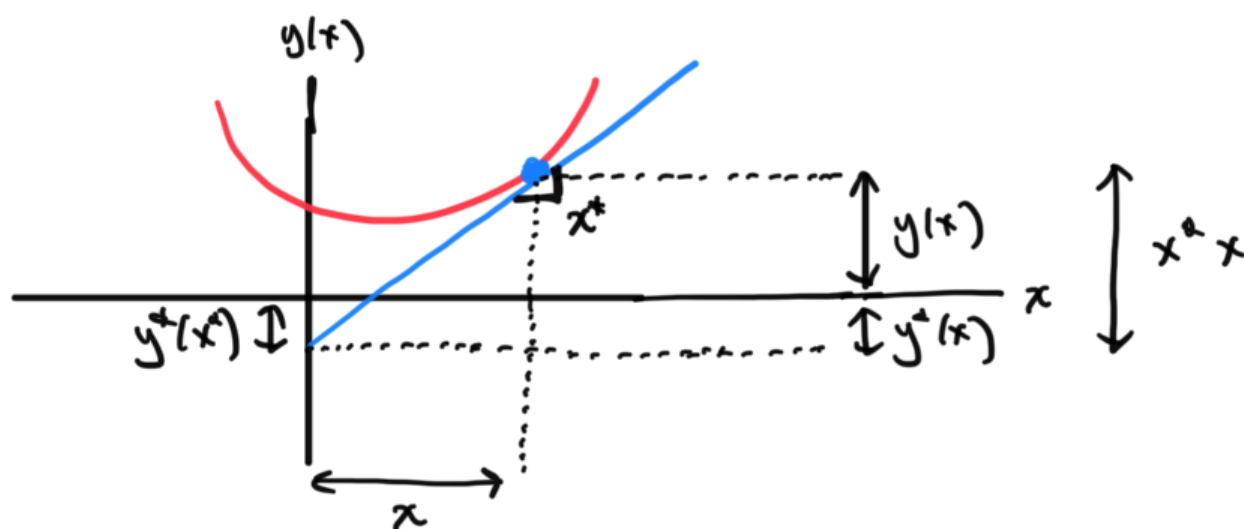


- $\Rightarrow x^*(x)$ can be inverted to obtain $x(x^*)$

- ... Legendre dual function $y^*(x^*)$ that encodes

- Can use this to derive ...
some info as $y(x)$

Graphical approach to Legendre transform



- 1) pick a point $(x, y(x))$
- 2) draw tangent line to $y(x)$ at x with slope $x^*(x)$
- 3) the tangent line has a y-intercept at $(0, -y^*(x^*))$
↗ def of $y^*(x^*)$

4) clearly:

$$y^*(x^*) + y(x) = x^* x$$

x^* and x are not indep:

linked through

$$x^*(x) = \left. \frac{\partial y}{\partial x} \right|_x$$

Comparison to stat mech:

dual variables

$$\begin{aligned} x^* &= \beta \\ y^* &= \beta F \\ y^*(x^*) &= \beta F(\beta) \end{aligned}$$

$$x^* = \frac{\partial y}{\partial x} \quad \beta = \frac{\partial S}{\partial E}$$

$$\begin{aligned} x &\sim \varepsilon \\ y &\sim S \\ y(x) &\sim S(\varepsilon) \end{aligned}$$

original variables

$$\begin{aligned} y^*(x^*) + y(x) &= x^* x \\ \beta F(\beta) + S(E) &= \beta E \end{aligned}$$

dual relation

divide by β :

$$\begin{aligned} F(kT) &= E - kT S(E) \\ \text{where } \frac{1}{kT} &= \frac{\partial S}{\partial E} \end{aligned}$$

traditional stat
Mech textbooks

Now with formalism... now specific examples

Simplest example: A single spin

$$\begin{array}{l} \uparrow \quad s = +1 \\ \downarrow \quad s = -1 \end{array} \quad E = -hs$$

$$Z(\beta, h) = \sum_{s=\pm 1} e^{+\beta h s} = e^{\beta h} + e^{-\beta h} = 2 \cosh \beta h$$

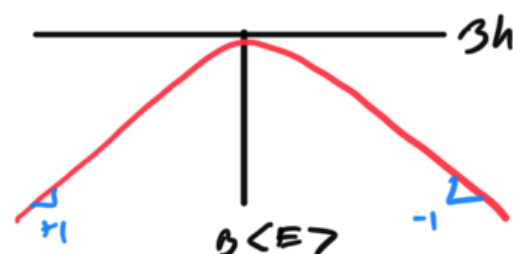
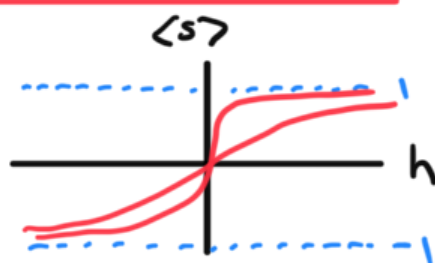
$$\cosh x \equiv \frac{e^x + e^{-x}}{2}$$

$$F(\beta, h) = -\frac{1}{\beta} \ln 2 \cosh(\beta h)$$

$$\langle E \rangle = -\frac{\partial}{\partial \beta} (\beta F) = \frac{\partial}{\partial \beta} [\ln 2 \cosh(\beta h)]$$

$$= h \frac{2 \sinh(\beta h)}{2 \cosh(\beta h)} = h \tanh(\beta h) = \langle E \rangle$$

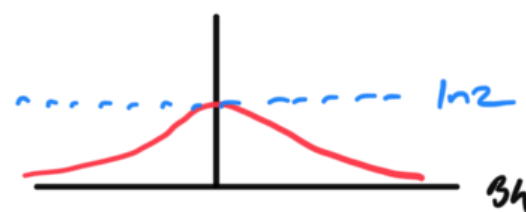
$$\langle E \rangle = h \langle s \rangle \Rightarrow \langle s \rangle = \tanh(\beta h)$$



$$F = \langle E \rangle - \frac{1}{\beta} \langle s \rangle$$

$$S(\beta) = \beta [\langle E \rangle - F(\beta)]$$

$$= \beta [\ln 2 \cosh(\beta h) + \beta h \tanh \beta h]$$



Example: our first interacting system Mean field ferromagnet

$$E(\vec{S}) = -\frac{1}{2N} \sum_{i,j=1}^N J s_i s_j - h \sum_{i=1}^N s_i$$

$$\begin{array}{ccc} \uparrow +J & \uparrow \downarrow & \uparrow \uparrow \\ \downarrow -J & \uparrow \uparrow & \downarrow \downarrow \end{array}$$

set $J=1$

$$\text{let } M = \frac{1}{N} \sum_{i=1}^N s_i \quad \text{simple mean magnetization}$$

$$\begin{aligned} \text{note } M^2 &= \left[\frac{1}{N} \sum_{i=1}^N s_i \right] \left[\frac{1}{N} \sum_{j=1}^N s_j \right] \\ &= \frac{1}{N^2} \sum_{i,j=1}^N s_i s_j \end{aligned}$$

$$E(\vec{S}) = -\frac{1}{2} N M^2 - N h M$$

energy of microstate $\vec{S}$ depends on $\vec{S}$!!

So $E(S) = 2 \dots$

only through Macroscopic variable

$$P(\vec{S}) \propto e^{-\beta E(\vec{S})} \Rightarrow P(m) \propto e^{-\beta E(m)} \underbrace{e^{S(m)}}_{\text{\# microstates with magnetization } m \dots}$$

$$\left. \begin{array}{l} \text{if } pN \text{ spins are up} \\ \text{and } (1-p)N \text{ spins are down} \end{array} \right\} m = p - (1-p) = 2p - 1 = m$$

$$p = \frac{m+1}{2}$$

So # of configurations w/ magnetization m = # of confs w/ frac $p = \frac{m+1}{2}$ spins up

we know this is $e^{NS(p)}$ w/ $S(p) = -p \ln p - (1-p) \ln (1-p)$

$$p = \frac{1+m}{2}$$

$$1-p = \frac{1-m}{2}$$

$$S(m) = -\frac{1+m}{2} \ln \frac{1+m}{2} - \frac{1-m}{2} \ln \frac{1-m}{2}$$

Summary:

$$P(m) \propto e^{-\beta \underbrace{[E(m) - kT S(m)]}_{F(m) \text{ restricted free energy}}}$$

$$P(m) \propto e^{-N\beta \underbrace{\left[\left(-\frac{m^2}{2} - hm \right) - kT S(m) \right]}_{F(m) \text{ free energy density}}}$$

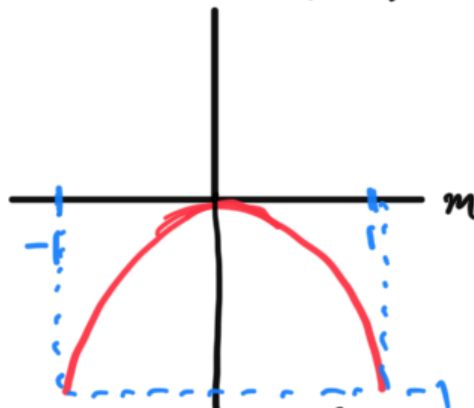
log dev prob for m

$$F(m) = \mathcal{E}(m) - kT S(m)$$

$$\mathcal{E}(m) = -\frac{m^2}{2} - hm$$

$\mathcal{E}(m)$ w/ $h=0$

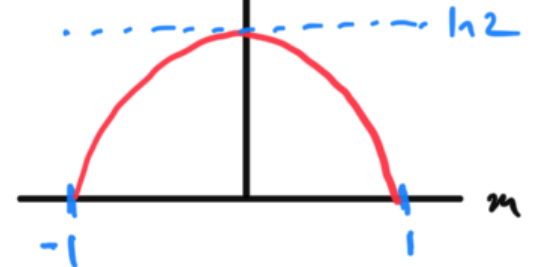
when $h=0$



Minimize energy $\mathcal{E}(m)$

send $m \rightarrow \pm 1$

$S(m)$



Maximize entropy $S(m)$

send $m \rightarrow 0$

send $m \rightarrow 0$

the most likely value of m

is $m^* = \arg \min F(m)$

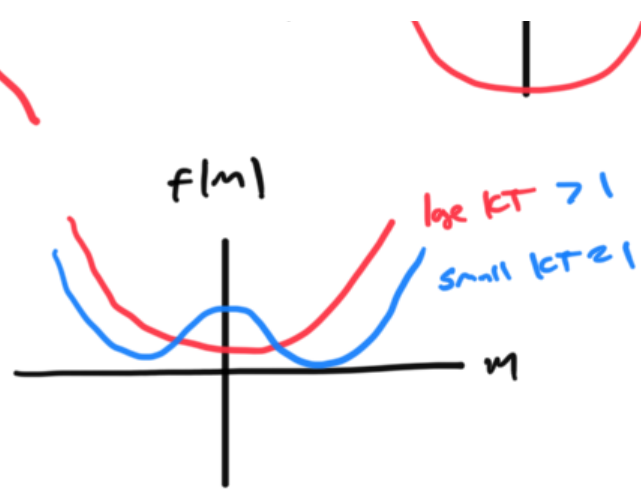
arises as a competition between

$\mathcal{E}(m)$



$-S(m)$

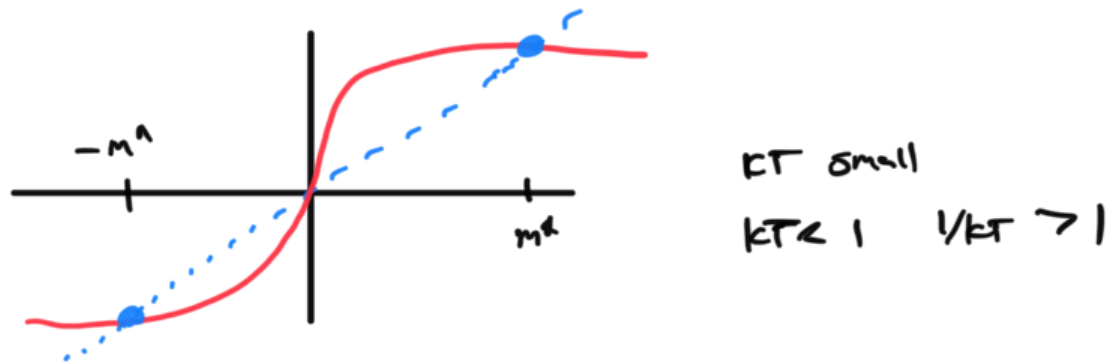
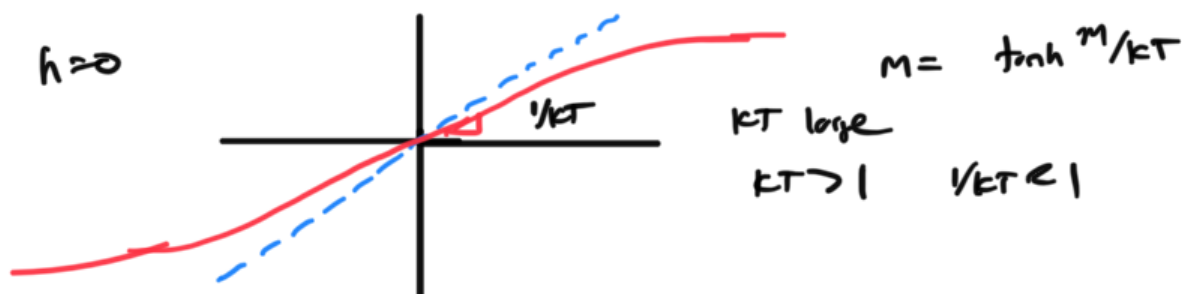




$$\frac{\partial f}{\partial m} = 0 \Rightarrow \frac{1}{2} \log \frac{1+m}{1-m} = \beta(m+h)$$

$$\tanh^{-1} x = \frac{1}{2} \log \frac{1+x}{1-x}$$

$$m = \tanh \beta(m+h)$$



heuristic derivation of $m = \tanh \beta m$



$$h=m \rightarrow \langle S \rangle = m$$

$$E = -\frac{1}{N} \sum_{i,j} S_i S_j$$

$$= \sum_i S_i m$$

- Each spin behaves like it feels a mean field $h=m$
- we know a single spin S in field h has magnetization $m = \tanh \beta h$
- self consistency demands that

$$m = \frac{1}{N} \sum_{i=1}^N \langle S_i \rangle = \frac{1}{N} \sum_{i=1}^N \tanh \beta m = \tanh \beta m$$